1. Objective:

Understand the mode of operation of monostable and astable multivibrators.

2. Preparation:

2.1. Repeat the mode of operation of monostable and astable multivibrators.
Especially using the NE555. See:

[1] Thomas Floyd “Digital Fundamentals”, ninth edition, Pearson Prentice Hall, 2006, chapter 7-6.

[2] Ulrich Tietze, Christoph Schenk “Halbleiterschaltungstechnik”, 11. Auflage, Springer, 1999, Kapitel 6.5.

2.2. Perform the calculations requested in 3.1.1 and 3.2.1.

3. Tasks:

3.1. <u>Monostable Multivibrator:</u>

Build a monostable multivibrator on the experimentation board using a NE555 integrated circuit (IC). To activate the trigger input (pin 2) use a push-button. See the last chapter (notes) on how to connect the push-button to the IC. Use C = 1µF and R_1 = 1MΩ. Use 10V as supply voltage for the circuitry.

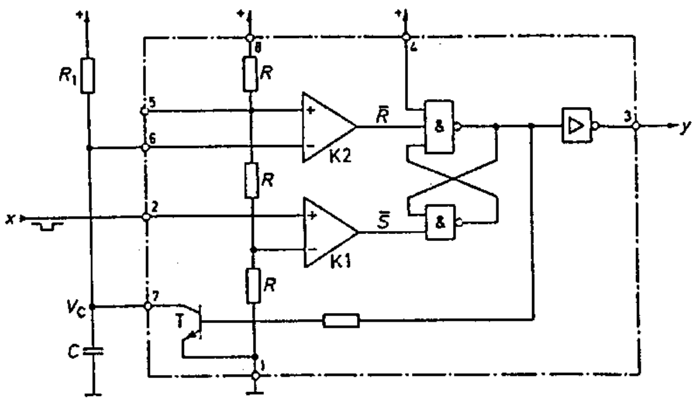
3.1.1. Calculate the pulse duration of the output signal “y” that should result.

3.1.2. Measure both the pulse duration at the output (pin 3) and the voltage across the capacitor C using the oscilloscope and its probes.

3.1.3. Explain possible differences between the calculated and the measured value.

3.1.4. Repeat the measurement from 3.1.2 using simple BNC-cables between the board and the oscilloscope. Explain the behavior.

3.1.5. Repeat the measurement from 3.1.2, measuring the output only, (not the capacitor).
Explain the differences to 3.1.2.



3.2. <u>Astable Multivibrator:</u>

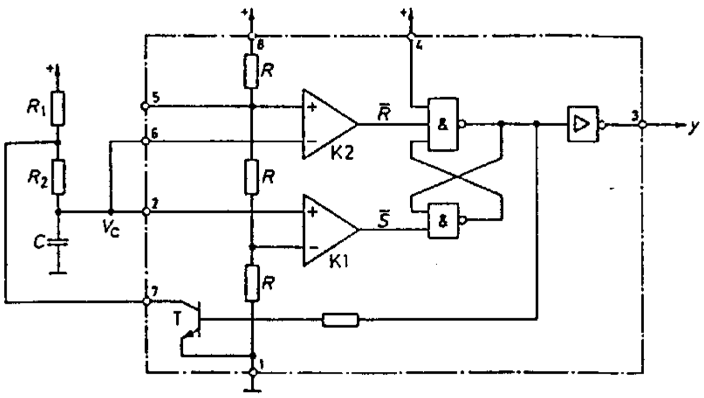
Build an astable multivibrator using the NE555. Use R_1 = 10kΩ, R_2 = 10 kΩ and C = 100nF. Use 10V as supply voltage for the circuitry

3.2.1. Calculate the vibration period that should result.

3.2.2. Measure both the pulse duration at the output (pin 3) and the voltage across the capacitor C using the oscilloscope and its probes.

3.2.3. Explain possible differences between the calculated and the measured value.

3.2.4. Zoom into one rising edge of the output and observe the waveform. Explain the behavior.



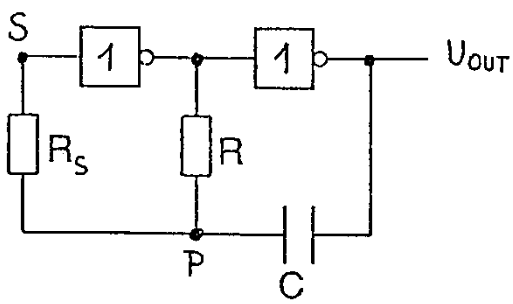
3.3. <u>Astable Multivibrator Using Two Inverter</u>

Build the following circuitry. Use the CMOS integrated circuit 74C00 and R_S = 1KΩ, R = 10kΩ, C = 1nF. Use 10V as supply voltage for the circuitry.

1.1. Understand the function of the circuit by visualizing the voltage at P and at the output on the oscilloscope.

1.2. Measure the voltage at S and P using the oscilloscope. Explain the curves.

1.3. What is R_S needed for?



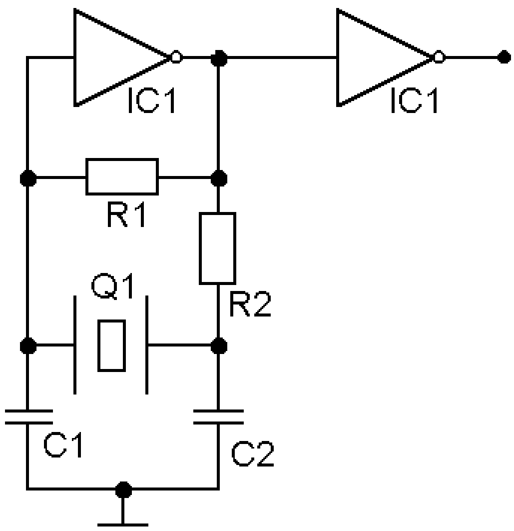
3.4. <u>Quartz Oscillator</u>

Build the following circuitry. Use the CMOS integrated circuit 74C00, a 12 MHz quartz crystal and R_1 = 1MΩ, R_2 = 220Ω, C_1 = C_2 = 68pF. Use **5V** as supply voltage for the circuitry.

1.1. Measure the output voltage using the oscilloscope and its probes.

1.2. Explain the curve.

1.3. What happens if you increase the supply voltage to 10V and to 15V?



4. Report:

4.1. The report should comply with C. Thorrold “Guide for writing lab reports”, Version 1.0, or another guide used in a previous semester (please state which).

4.2. Special focus is needed on logically clear explanation of the observed behavior.

5. Notes:

In order to add a switch or a push-button to a circuit, the following schematic can be used:

If the switch is open: x_i will have a “high” level
If the switch is closed: x_i will have a “low” level